

# Medical Image Segmentation with U-Net and Vision Transformers: A Comparative Study on ISIC 2018

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**Abstract**—Automated skin lesion segmentation is a critical prerequisite for computer-aided diagnosis of melanoma. While Convolutional Neural Networks (CNNs), particularly the U-Net architecture, have established strong baselines for this task, they are fundamentally limited by local receptive fields. Recently, Vision Transformers (ViTs) have demonstrated strong capability in modeling long-range global context through self-attention mechanisms. In this work, we present a comparative analysis between a purely convolutional U-Net trained from scratch and a hybrid CNN-Transformer architecture (TransUNet) initialized with ImageNet-21k pre-trained weights. Both architectures were trained and evaluated on the ISIC 2018 skin lesion segmentation dataset under constrained hardware limits (15 GB VRAM). Experimental results demonstrate that naive fine-tuning of transformer-based encoders causes optimization instability and degraded generalization. To address this, a two-stage progressive unfreezing strategy with differential learning rates was implemented. The optimized TransUNet achieved a Dice Similarity Coefficient (DSC) of 0.9139 and Recall of 0.9043, outperforming the U-Net baseline which achieved a DSC of 0.9014. Although TransUNet produced superior segmentation accuracy and sensitivity, this improvement required significantly higher computational complexity and parameter count. The findings indicate that transformer-enhanced segmentation architectures can provide measurable benefits for medical image analysis when carefully fine-tuned under limited data conditions.

**Index Terms**—Deep Learning, Medical Image Segmentation, U-Net, TransUNet, Vision Transformers, Melanoma Detection, ISIC 2018.

## I. INTRODUCTION

Skin cancer is among the most prevalent forms of cancer worldwide, with melanoma representing one of the deadliest variants due to its high metastatic potential. Early diagnosis significantly increases survival probability, motivating the development of computer-aided diagnostic systems for dermatological analysis.

Dermoscopic imaging provides high-resolution visualization of skin lesions; however, accurate manual segmentation remains labor-intensive, subjective, and highly dependent on clinical expertise. Automated segmentation systems are therefore essential for assisting clinicians in lesion boundary extraction and quantitative analysis.

Medical image segmentation presents several challenges including low contrast boundaries, lesion variability, artifacts such as hair occlusions and air bubbles, illumination inconsistency, and irregular lesion morphology. Deep learning methods, particularly Fully Convolutional Networks (FCNs), have achieved significant progress in this domain.

U-Net, introduced by Ronneberger *et al.* [1], became the standard architecture for biomedical image segmentation due to its encoder-decoder structure and skip connections that preserve spatial information. Despite its effectiveness, convolution operations inherently possess local receptive fields, limiting their ability to model long-range contextual dependencies.

Recently, Vision Transformers (ViTs) [4] have emerged as an alternative paradigm capable of learning global relationships using self-attention mechanisms. TransUNet [2] combines CNN-based feature extraction with transformer encoders to capture both local and global semantic information.

This study investigates whether transformer-enhanced segmentation architectures provide meaningful performance improvements for skin lesion segmentation on the ISIC 2018 dataset under practical hardware constraints. Specifically, we compare a conventional U-Net against TransUNet and analyze the trade-off between segmentation performance and computational complexity.

## II. RELATED WORK

### A. CNN-Based Medical Segmentation

Convolutional Neural Networks have dominated medical image segmentation for nearly a decade. U-Net [1] introduced a symmetric encoder-decoder structure with skip connections that enabled precise localization while preserving contextual information. Due to its effectiveness with limited training data, U-Net became the foundational architecture for numerous segmentation systems.

Several extensions of U-Net have since been proposed. Attention U-Net introduced attention gates to improve focus on salient anatomical regions, while Residual U-Net incorporated residual blocks to stabilize deeper networks. nnU-Net [3] further demonstrated that carefully optimized training

heuristics and preprocessing pipelines are often as important as architectural novelty.

Despite strong performance, CNNs remain fundamentally constrained by local convolution kernels. Capturing global spatial relationships requires increasingly deep architectures, which raises computational complexity and optimization difficulty.

### B. Transformers in Computer Vision

Transformers were originally introduced for Natural Language Processing tasks and later adapted to computer vision through the Vision Transformer (ViT) architecture [4]. ViTs divide images into patches and process them as token sequences using self-attention mechanisms capable of modeling long-range dependencies.

Transformer-based architectures rapidly gained popularity in medical imaging. Chen *et al.* proposed TransUNet [2], which integrates a ViT encoder into a U-Net-like framework to combine global contextual modeling with precise localization. Swin-Unet [5] further extended this concept through hierarchical shifted-window attention mechanisms.

Although transformer-based models have demonstrated superior performance on several segmentation benchmarks, they typically require larger datasets, stronger regularization, and more careful optimization compared to CNNs. Their effectiveness for noisy 2D dermatological datasets under limited hardware conditions remains an active area of investigation.

## III. METHODOLOGY

### A. Dataset and Preprocessing

Experiments were conducted using the ISIC 2018 Task 1 skin lesion segmentation dataset [6]. The dataset contains 2,594 dermoscopic training images with corresponding pixel-level segmentation masks.

A 90/10 split was used for training and validation. All images were resized to  $256 \times 256$  pixels to maintain consistent input dimensions and reduce GPU memory consumption.

To improve generalization and reduce overfitting, extensive augmentation was applied using the Albumentations framework. Augmentation operations included:

- Horizontal and vertical flipping
- Random rotations
- Affine transformations
- Gaussian blur
- Color jitter
- CoarseDropout regularization

All images were cached into RAM using a multi-threaded preprocessing pipeline implemented with `ThreadPoolExecutor` to minimize storage I/O bottlenecks during training.

### B. Baseline U-Net Architecture

The baseline model was implemented using a conventional U-Net encoder-decoder structure. The encoder consisted of four downsampling stages with double  $3 \times 3$  convolutions followed by Batch Normalization and ReLU activations.

Feature dimensions increased progressively from 64 to 512 channels at the bottleneck. The decoder utilized bilinear upsampling followed by feature concatenation through skip connections.

The architecture contained approximately 31 million trainable parameters.

### C. TransUNet Architecture

The transformer-based model followed the TransUNet design proposed in [2]. A CNN encoder first extracted spatial feature maps from the input image. These features were projected to 768 dimensions using a  $1 \times 1$  convolution before tokenization.

The resulting sequence of tokens was processed through 12 Vision Transformer layers using multi-head self-attention. Positional embeddings were incorporated to preserve spatial ordering.

Transformer outputs were reshaped into 2D feature maps and passed into a decoder structurally similar to U-Net.

The complete architecture contained approximately 105 million parameters.

### D. Loss Function

Medical segmentation datasets often exhibit severe class imbalance between foreground lesions and background pixels. To address this issue, a hybrid Binary Cross Entropy (BCE) and Dice Loss objective was used:

$$\mathcal{L}_{total} = \mathcal{L}_{BCE}(y, \hat{y}) + \left[ 1 - \frac{2 \sum (y \cdot \hat{y}) + \epsilon}{\sum y + \sum \hat{y} + \epsilon} \right] \quad (1)$$

This combined objective stabilizes optimization while directly optimizing segmentation overlap quality.

## IV. TRAINING STRATEGY

Training transformer-based architectures on limited medical datasets is challenging because transformers generally lack the strong inductive biases of CNNs and are more susceptible to overfitting.

To improve convergence stability, the ViT encoder was initialized using ImageNet-21k pre-trained weights.

Initial experiments revealed that directly fine-tuning the entire TransUNet architecture with a uniform learning rate caused unstable optimization and degraded validation performance. This behavior is attributed to destructive updates applied to the pre-trained transformer representations.

To mitigate this issue, a progressive unfreezing strategy was implemented:

- 1) **Phase 1 (Epochs 0–4):** Transformer and CNN encoder layers remained frozen while only decoder layers were trained.
- 2) **Phase 2 (Epochs 5–50):** The entire network was unfrozen with differential learning rates.

A learning rate of  $1 \times 10^{-5}$  was applied to pre-trained transformer layers, while decoder layers used  $1 \times 10^{-4}$ .

Both models were trained for 50 epochs using the Adam optimizer with cosine annealing learning rate scheduling.

Training utilized mixed precision FP16 computation on an NVIDIA Tesla T4 GPU within Google Colab. Batch size was restricted to 8 samples due to VRAM limitations.

## V. EXPERIMENTS AND RESULTS

### A. Evaluation Metrics

Model performance was evaluated using the following metrics:

- Dice Similarity Coefficient (DSC)
- Intersection over Union (IoU)
- Recall (Sensitivity)
- Specificity

Dice Similarity Coefficient was selected as the primary metric because it directly measures overlap quality between predicted and ground-truth masks.

### B. Initial Baseline Comparison

An initial 30-epoch experiment using naive fine-tuning with a uniform learning rate of  $10^{-4}$  resulted in unstable transformer optimization. Under this configuration, TransUNet achieved a DSC of 0.8858, underperforming compared to the baseline U-Net which achieved 0.8978 DSC.

These findings highlighted the importance of careful optimization scheduling for transformer-based medical segmentation models.

### C. Optimized 50-Epoch Evaluation

After implementing progressive unfreezing and differential learning rates, TransUNet demonstrated significantly improved generalization performance.

TABLE I: Validation Segmentation Metrics

| Model                  | DSC           | IoU           | Recall        | Specificity   |
|------------------------|---------------|---------------|---------------|---------------|
| U-Net                  | 0.9014        | 0.8206        | 0.8830        | 0.9763        |
| TransUNet (Frozen)     | 0.9093        | 0.8337        | 0.8842        | <b>0.9811</b> |
| TransUNet (Fine-Tuned) | <b>0.9139</b> | <b>0.8415</b> | <b>0.9043</b> | 0.9767        |

The optimized TransUNet achieved the best Dice score of 0.9139, surpassing the baseline U-Net by approximately 1.25%.

A particularly important improvement was observed in Recall, where TransUNet improved sensitivity from 0.8830 to 0.9043. This indicates superior detection of lesion regions and reduced false negatives, which is highly desirable in melanoma screening systems.

### D. Computational Analysis

Although TransUNet achieved superior segmentation performance, the improvement came at substantial computational cost.

TABLE II: Model Complexity Comparison

| Model     | Parameters | Relative Complexity |
|-----------|------------|---------------------|
| U-Net     | 31M        | 1×                  |
| TransUNet | 105M       | 3.39×               |

The transformer-based model required significantly more memory, longer training time, and more complex hyperparameter tuning.

## VI. DISCUSSION

The experiments demonstrate that integrating transformer encoders into medical segmentation architectures can provide measurable improvements in segmentation quality, particularly for difficult lesion boundaries.

The global self-attention mechanism of Vision Transformers appears effective in modeling long-range contextual relationships that conventional CNNs struggle to capture.

However, the results also emphasize several practical limitations of transformer-based segmentation systems:

- Increased parameter count
- Higher GPU memory consumption
- Greater optimization sensitivity
- Longer training times
- Strong dependence on pre-training

For high-performance clinical servers or cloud-based analysis pipelines, TransUNet provides meaningful advantages in sensitivity and segmentation accuracy.

However, for mobile or resource-constrained deployment scenarios, the baseline U-Net remains highly competitive due to its significantly lower computational overhead while still achieving over 0.90 DSC.

## VII. CONCLUSION

This work presented a comparative study between a conventional U-Net and a transformer-enhanced TransUNet architecture for skin lesion segmentation on the ISIC 2018 dataset.

Experimental results showed that naive transformer fine-tuning can lead to unstable optimization and reduced performance. By introducing a progressive unfreezing strategy with differential learning rates, the transformer-based model achieved superior segmentation accuracy and sensitivity.

The optimized TransUNet achieved a Dice Similarity Coefficient of 0.9139, outperforming the baseline U-Net score of 0.9014.

Although transformer architectures provide improved global contextual modeling, these gains require substantially greater computational resources and careful optimization.

Future work may explore lightweight transformer variants, hybrid attention modules, and knowledge distillation approaches to improve deployment efficiency while preserving segmentation quality.

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